

Protective effect of ATP in experimental hemorrhagic shock

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The essential role of adenosine triphosphate (ATP) in energy metabolism has resulted in numerous studies of its possible significance in shock. Due to curious circumstances, however, its vasodilatory and hypotensive effects have attracted more experimental attention than its possible therapeutic role in shock. Recently, Talaat and co-workers in a series of studies²³⁻²⁶ have suggested that ATP may be protective to the shocked animal. This is a study of the effect of ATP on rats and dogs in hemorrhagic shock, designed to determine whether ATP is of prophylactic or therapeutic benefit, and to investigate the possible mechanism of such activity.

MATERIALS AND METHODS

Part I. ATP prior to hemorrhage. Adult (300 to 350 Gm.) Sprague-Dawley rats were lightly narcotized with 0.1 mg. per 100 Gm. of body weight of Nembutal administered intraperitoneally in 0.1 ml. of saline. When drowsy, the animals were restrained on a horizontal board and the left carotid artery was cannulated with a 0.032 mm. i.d. polyethylene catheter utilizing 0.5 ml. of 0.5 percent Xylocaine local anesthesia. No anticoagulants were employed and the animals remained sufficiently alert to respond to normal stimuli. All animals had been fasting

for 18 hours prior to the experiment, but had been allowed water ad lib.

Fifteen minutes after the intraperitoneal administration of 10 mg. of ATP, or an equivalent amount of the various control drugs, the animals were bled, 2.25 ml. per 100 cm.² of body surface, calculated according to the method of Sayers and co-workers²¹ and Lea and Clarke.¹³ Bleeding was carried out in six equal amounts each 5 minutes during a 30 minute period in order to avoid the complication of a more gradual hemorrhage that invariably confuses evaluation of vasodilatory drugs in hemorrhagic shock.

Arterial blood pressure was recorded on a mercury manometer connected to the carotid cannula. At the completion of bleeding the cannula was removed, the artery ligated, and the wound closed.

Both ATP and adenosine monophosphate (AMP) were stored frozen in crystalline state. Fresh solutions of 100 mg. per milliliter were prepared immediately prior to use in each experiment.

In order to compare equivalent vasodilatory effects of ATP, AMP, and Dibenzylamine (phenoxybenzene hydrochloride, DBZ), preliminary experiments were performed on 15 rats, recording the arterial pressure for 4 hours following the intraperitoneal administration of various doses of ATP, AMP, and DBZ. The final results of these studies are illustrated in Fig. 1. It was decided to consider that 10 mg. of ATP was the equivalent of 16 mg. of AMP and of 1 mg. of intravenously administered Dibenzylamine. In order

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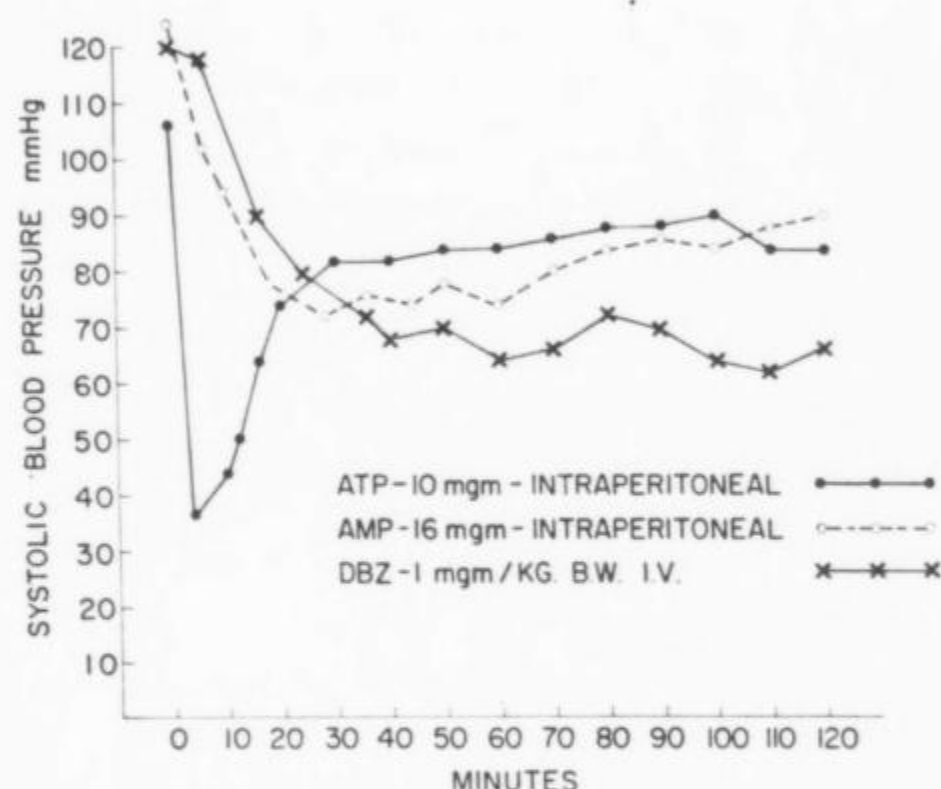


Fig. 1. Hypotensive effect of ATP, AMP, and DBZ.

to compare the possible activity of pyrophosphate in its effect on hemorrhagic shock, the equivalent dose on a weight basis (1 mg.) of this substance was administered in a series of 20 animals. This is the maximum solubility of this compound at 20° C.

All drugs were administered in 0.1 ml. of saline. Controls received saline alone.

Although the majority of deaths occurred within 12 hours following hemorrhage, survivors were observed for 48 hours. All animals were allowed food and water ad lib following hemorrhage.

A total of 110 animals were thus tested. Of these, 30 received ATP; 20 saline alone; 20 AMP; 20 Dibenzyline; and 20 pyrophosphate. In one simultaneous series of 20 animals a double blind study was performed with the individual bleeding the animals unaware of whether prior treatment had been with saline or ATP. In other studies the choice of treatment was by random selection.

Part II. ATP following hemorrhage. In a total of 20 animals, the protocol for bleeding was carried out as outlined above, except that the drug was administered 15 minutes following the termination of bleeding. In this study, 20 animals received ATP following hemorrhage.

As in Part I, the animals were observed for 48 hours, even though essentially the

entire mortality occurred in the first 2 hours following bleeding and drug administration.

Part III. ATP content of heart and liver. These experiments were performed to determine the concentration of ATP in the heart and in the liver, both in the control shocked animals and in shocked animals after the administration of ATP.

The ATP concentration in heart and liver was studied in a series of 20 rats given 10 mg. of ATP prior to bleeding, as described in Part I. Thirty minutes after the termination of bleeding, each animal was quickly killed by excising either the heart or the liver via a thoracotomy or laparotomy. The appropriate organ was quickly drained of blood and dropped into a mixture of acetone and dry ice. Less than 5 seconds characteristically elapsed between the incision and immersion into the freezing solution. When frozen, the organ was weighed and immediately homogenized with 5.0 ml. of 6 percent trichloroacetic acid, with the homogenizer immersed in ice. The homogenate was centrifuged and the supernatant analyzed for its ATP content according to the enzymatic method of Holloway,¹¹ Dianzani,⁴ and Bucher.²

Part IV. ATP effect on cardiac function of dogs in hemorrhagic shock. In order to investigate the effect of ATP on cardiac function of the shocked animal, a series of 26 dogs were studied, utilizing the Wiggers technique.^{5, 12, 29}

Adult mongrel dogs were anesthetized with 6 mg. per kilogram of veterinarian Nembutal, were intubated, and underwent left thoracotomy through the 4th interspace. Cardiac output was measured by placing a "C" probe from an electromagnetic flowmeter around the descending aorta immediately distal to the subclavian artery. The thoracotomy wound was closed and the dog left in the right lateral position. A rectal temperature probe and a heating pad maintained body temperature at 37° C. The animal was allowed to awaken and thereafter kept only drowsy enough to avoid expulsion of the endotracheal tube.

The left femoral artery was cannulated and utilized for the electronically recorded blood pressure.

A large bore cannula was placed in the right femoral artery and the animal was given 3 mg. per kilogram of body weight of heparin and one million units of penicillin. The femoral artery cannula led via a large bore cannula to a clean, siliconized, but not sterile Lamson bottle. The animal was allowed quickly to bleed into the bottle held at body level, and characteristically settled at a systolic arterial pressure of 50 mm. Hg within 2 minutes. The bottle was elevated approximately 45 cm. above heart level, the exact height being adjusted to maintain a systolic pressure of 50 mm. Hg. Within 5 minutes equilibrium was established. A point on the bottle was marked recording 80 percent of the blood thus shed into the reservoir.

(a) *ATP with total blood replacement in shock.* Characteristically, the animals began to take up blood from the reservoir at the end of 2 to 4 hours of hypotension at 40 mm. Hg. Such autotransfusion from the reservoir coincides with the fall in peripheral vascular resistance and classically is supposed to announce "irreversible shock." In four animals 6.66 mg. of ATP per kilogram of body weight were given when 20 percent of the total shed blood in the reservoir was thus taken up. Simultaneous with ATP administration, the remaining 80 percent of shed blood was reinfused. In four control animals the same protocol was employed, but ATP omitted. Blood pressure and cardiac output were followed for 3 hours following blood replacement.

This experiment simulated the administration of ATP with simultaneous adequate blood replacement to a patient in hemorrhagic shock.

(b) *ATP without blood replacement in profound shock.* In six experiments an identical protocol was employed, except that at the point of 20 percent blood take-up and ATP administration the cannula to the reservoir was clamped. This left the animal in an 80 percent deficit of the shed blood. The

arterial pressure and cardiac output were followed for 2 hours thereafter.

This experiment simulated ATP administration to a deeply shocked subject with inadequate fluid replacement.

(c) *ATP alone in moderate shock.* In 12 animals hypotension to 50 mm. Hg was induced as before. Following 60 minutes of such hypotension, ATP, 6.66 mg. per kilogram of body weight, was administered as previously, but without further blood replacement. The reservoir cannula was clamped to avoid any further uptake of blood.

This experiment was designed to simulate ATP administration to a mildly shocked subject without adequate blood replacement.

RESULTS

Part I. ATP, AMP, DBZ, and PP prior to hemorrhage. The results of ATP and suitable control drug administration prior to hemorrhage are shown in Table I and illustrated graphically in Fig. 2. The mortality with saline alone was 70 percent in 20 rats so tested. There was no statistically significant difference between this mortality and the 75 percent mortality ($p = > 0.5$) with AMP, the 80 percent mortality ($p = > 0.5$) with Dibenzyline, or the 60 percent mortality ($p = > 0.1$) with pyrophosphate. The mortality of 20 percent in the 30 animals given ATP, however, was but 20 percent (6 of 30 animals). This was significant at a high level of expectancy ($p = < 0.01$) compared to saline controls.

Part II. ATP following hemorrhage. The survival of shocked rats given ATP 30 minutes following hemorrhage is shown in Fig. 3, as compared to saline controls and to animals given ATP prior to bleeding. Whereas the control had 70 percent mortality, the animals given ATP after bleeding had but 40 percent mortality, with 12 of the 20 rats receiving the drug thus surviving. This figure is only significant at a 5 percent value.

Part III. ATP contents of heart and liver. Table II provides the data pertaining to the ATP concentrations in the hearts and livers of 10 shocked rats given saline as a

Table I. Mortality following prophylactic drug administration

Drug	Dose	Total animals	Mortality (%)	Difference from saline control (P value)
Saline	0.1 ml.	20	70	
Adenosine monophosphate (AMP)	16 mg./Kg.	20	75	> 0.5
Dibenzylidine (DBZ)	1 mg./Kg.	20	80	> 0.5
Pyrophosphate (PP)	1.0 mg.	20	60	> 0.5
Adenosinetriphosphate (ATP)	10 mg./Kg.	30	20	< 0.001

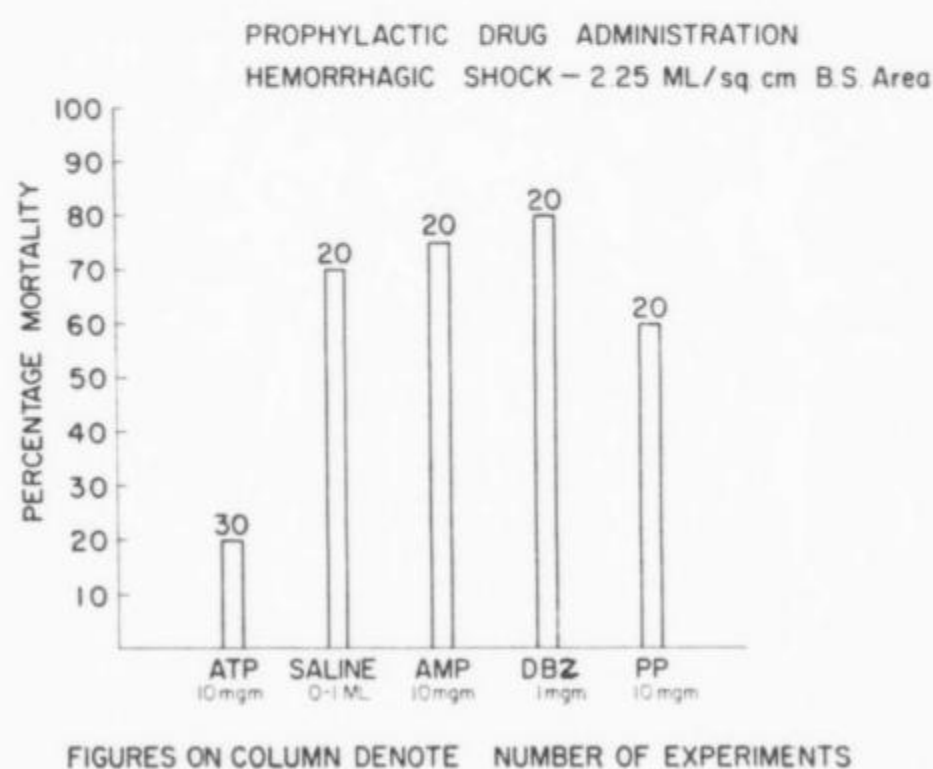


Fig. 2. Graph illustrating mortality following ATP, AMP, DBZ, and PP.

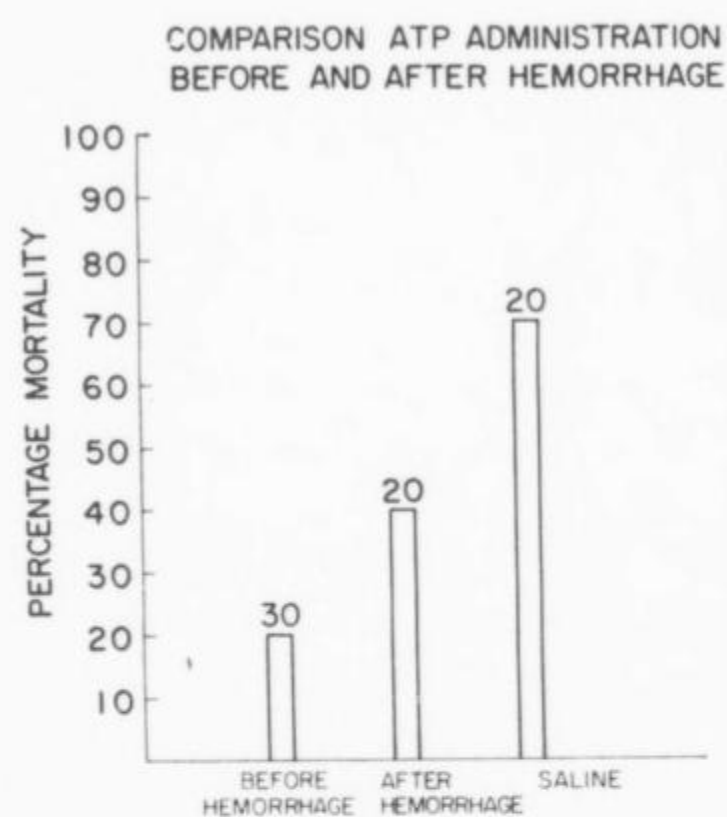


Fig. 3. Mortality following ATP administration to shocked rats.

control and an equal number given 10 mg. of ATP intraperitoneally immediately prior to bleeding. Fig. 4 graphically depicts these data.

(a) *Heart.* In keeping with the prior

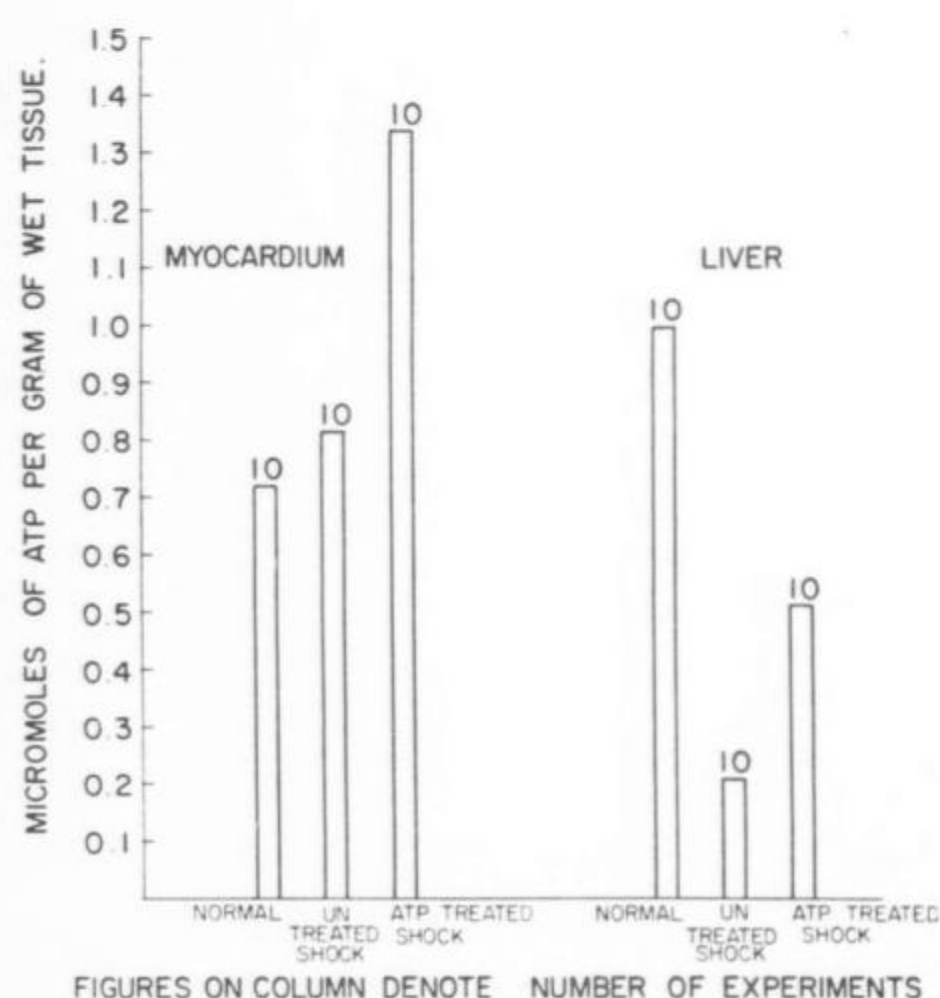
findings of others,^{14, 17, 18} the ATP content of the heart was not significantly depressed following hemorrhagic shock. In the normal, nonshocked controls, the mean ATP content of the heart was 0.72 μ moles per gram (S.D. = 0.12). In the untreated shocked animals the equivalent figure was 0.81 μ moles per gram (S.D. = 0.31), a difference that is not statistically significant. In the animals receiving prophylactic ATP, the cardiac ATP concentration was 1.34 μ moles per gram (S.D. = 0.48), which was significantly above the untreated control ($p = < 0.01$).

(b) *Liver.* The ATP concentration was 0.99 μ moles (S.D. = 0.28) in unshocked normals, compared to the saline treated shocked group with a mean hepatic concentration of 0.21 μ moles (S.D. = 0.28). This depletion of hepatic ATP following hemorrhagic shock is statistically significant ($p = < 0.01$) and is in keeping with the previous findings of others.¹⁴⁻¹⁶ In the animals given ATP prior to hemorrhage, however, there was a mean concentration of 0.511 μ moles per gram (S.D. = 0.2), which is a fivefold increase over the untreated shocked animals ($p = < 0.01$).

The tissue concentrations, as thus measured, represent a combination of drug absorption from the peritoneal cavity, its fixation by the tissue and the rate of its utilization. In these experiments, if all of the 10 mg. of intraperitoneally administered drug had been absorbed, had remained unchanged and spread evenly throughout the tissues of the 300 Gm. animal, one would anticipate a concentration of 0.033 mg. per gram of tissue. The method of measuring ATP differentiates

Table II. ATP concentration in liver and heart following hemorrhagic shock

	No. animals	ATP (micromoles per Gm. wet tissue)		
		Range	Mean conc.	Std. deviation
Heart				
Control	10	0.50 - 0.98	0.72	0.12
Untreated shock	10	0.59 - 0.86	0.81	0.31
ATP shock	10	0.74 - 2.34	1.34	0.48
Liver				
Control	10	0.52 - 1.41	0.986	0.28
Untreated shock	10	0.29 - 0.95	0.524	0.05
ATP shock	10	0.12 - 0.28	0.207	0.2



FIGURES ON COLUMN DENOTE NUMBER OF EXPERIMENTS

Fig. 4. Graph showing ATP concentration in heart and liver of shocked rats following ATP administration.

ATP from ADP and AMP, and it can be anticipated that a large percentage of the high energy ATP is dephosphorylated as soon as adsorbed, so that the value of ATP in both heart and liver represent 15 to 21 times the concentration that could be anticipated upon equal distribution alone. Concentration in the tissue would be inversely proportional to the rate of drug utilization.

The data of these experiments, however, demonstrate that ATP is being absorbed from the shocked animals and that it is being concentrated in the heart and in the liver of these animals compared to the de-

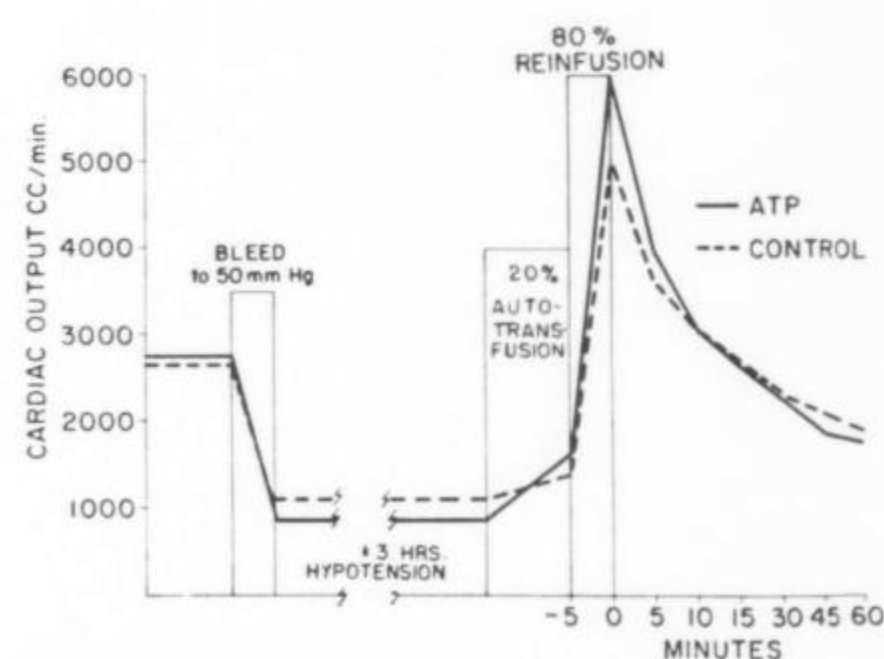


Fig. 5. Cardiac output and blood pressure in 4 dogs receiving ATP at time of total blood replacement and 4 controls in profound hemorrhagic shock.

pleted state of their nontreated and shocked controls.

Part IV. Cardiac output and blood pressure in dogs shocked utilizing the Wiggers technique and given ATP.

(a) *ATP with total blood replacement in profound shock.* Fig. 5 illustrates the mean cardiac output and blood pressure response in eight dogs, four of which had received ATP with reinfusion of the remaining 80 percent of the shed blood in the reservoir. There was no significant difference in the cardiac output of the dogs given ATP and their controls. In both, the cardiac output rose to near its normal level upon administration of the blood-ATP combination.

Blood pressure lagged in returning to normal in the ATP treated dogs due pre-

sumably to its vasodilatory effect. Within 5 minutes, however, the ATP animals and controls had essentially equal blood pressure.

(b) *ATP without blood replacement in profound shock.* ATP administered without further blood replacement when the animal had taken up 20 percent of its shed blood had no significant effect on cardiac output compared to controls. As shown in Fig. 6, there was a transient depression of blood pressure following drug administration with a concomitant fall in cardiac output, but within 5 minutes both were essentially at comparable levels of the controls.

(c) *ATP alone in moderate shock.* When administered alone 1 hour following the onset of hemorrhagic hypotension, ATP produced an immediate and further significant depression in both blood pressure and cardiac output, which characteristically lasted for 3 to 5 minutes and was followed by a slight rebound above pre-ATP administration levels. The rebound in cardiac output usually lasted less than 20 minutes, was normally of an almost imperceptible magnitude, and its total at best equaled the short but precipitous fall in output that occurred immediately following ATP administration. Fig. 7 records the mean values and standard deviations of the cardiac output and blood pressures in these studies. There was no total gain in cardiac output by ATP administration.

DISCUSSION

The intimate relation of ATP and energy transfer historically suggested that it might play an important role as the ephemeral "shock toxin,"⁸ since its administration produced both hypotension and vasodilatation.⁶ A number of studies,^{10, 19, 27} however, failed to identify high concentrations of ATP in blood or tissues of shocked animals.

Depletion of ATP in shocked animals was noted in 1945 by McShan and co-workers¹⁷ and quantitated more specifically a year later by LePage,¹⁴ who found tissue depletion in liver, kidney, brain, heart, and skeletal muscle, in that order of magnitude.

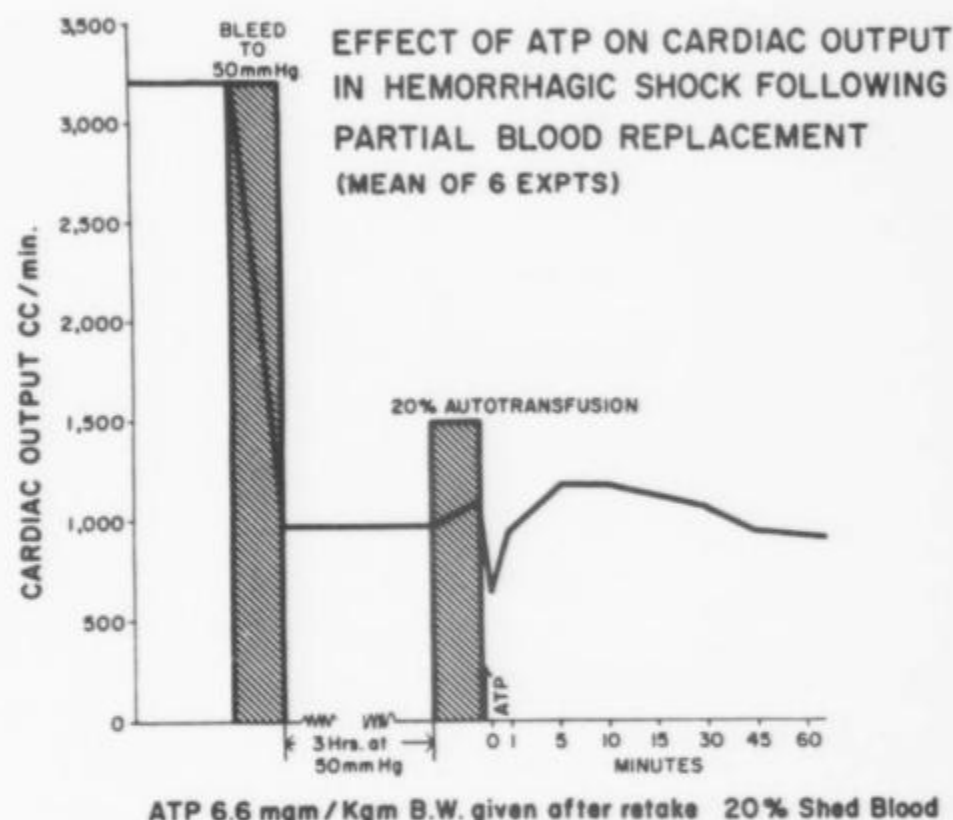


Fig. 6. Cardiac output and blood pressure in 6 dogs receiving ATP and 6 controls without total blood replacement in profound hemorrhagic shock.

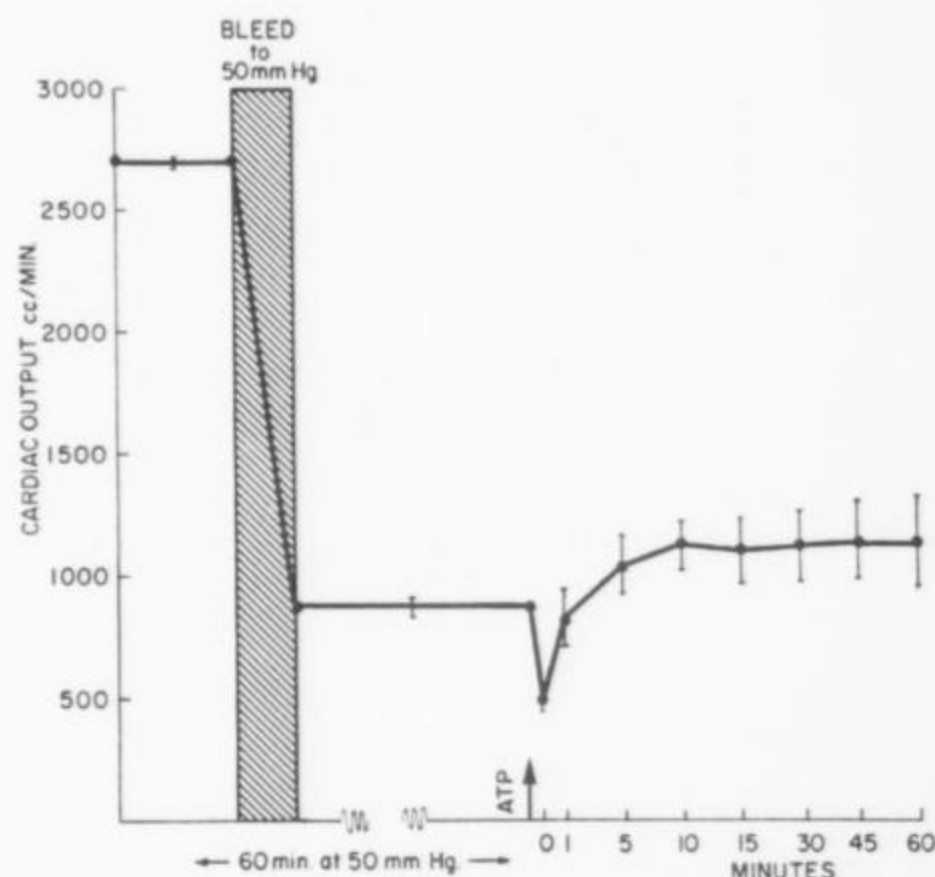


Fig. 7. Cardiac output and blood pressure in 4 dogs, half of which received ATP without blood replacement following 1 hr. of hypotension to 50 mm. Hg

Levenson and co-workers¹⁵ and Migone¹⁸ confirmed these findings, using specific enzymatic methods.

The details of the effects of ATP on both cardiac and peripheral vascular activities have been thoroughly investigated and more recently summarized by Rowe and co-workers.²⁰ They found an increased output of 25 percent, an increase in coronary flow of 534 percent, and an elevated ventricular oxygen consumption of 48 percent. They

and Green and Stoner⁷ conclude that the "ATP effect" on the heart results primarily from coronary vasoconstriction and peripheral venous dilatation.

With these extensive previous studies of ATP, it is curious that its possible therapeutic effect in shock has been so largely neglected except for a few isolated investigators.

In Japan, the ATP research council has stimulated evaluation of this drug in a variety of conditions and concludes that it is of probable therapeutic value in shock.¹

Talaat and co-workers, first in Cairo and later in Oklahoma City,²³⁻²⁶ have confirmed that 100 mg. of ATP administered to dogs before hemorrhage will increase survival from 20 percent in controls to 80 percent in the ATP group. When administered after bleeding, there was 55 percent survival. The heterogeneity of the dog population in their studies suggested these analogous experiments in rats.

The current studies support the thesis of Talaat and his co-workers²³⁻²⁶ that ATP is protective if given prior to bleeding, but not if administered thereafter. The possible mechanism of such protection is obscure. The known vasodilatory effect of ATP suggests this mechanism, but the current experiments obviate this possibility since more potent vasodilatory agents, such as dibenzylamine or hypotensive equivalent doses of AMP or ADP, were not similarly protective. Since none of the control experiments utilizing pyrophosphate, AMP, and ADP showed the beneficial effect demonstrated by ATP, it can be assumed that its pharmacologic activity probably resides in the high-energy phosphate bond of the terminal phosphate, which is unique to ATP.

Depression in ATP content of the liver, and to a much lesser extent in the heart, following hemorrhage^{14, 17, 18} suggests that ATP administration might provide a beneficial compensation for a deficit in these organs provoked by bleeding. Crowell and Guyton³ believe that myocardial failure is the lesion primarily responsible for death from oligemic hypotension. If this be so, it

is reasonable to hypothesize that a discrepancy in high energy phosphate bonds might be the ultimate mechanism thus interfering with visceral function. There was, however, no significant depression of myocardial ATP following bleeding. Neither was there evidence that ATP improved the depressed cardiac output occurring during oligemic hypotension. It is possible, but by no means certain, that the increased hepatic concentration of ATP was protective.

It is, indeed, difficult to understand how ATP is protective to the shocked animals. Ten milligrams of ATP administered intraperitoneally must be absorbed and inactivated relatively quickly in an animal otherwise deprived of such high energy phosphate bonds. The fact that the ATP administration given prior to bleeding results in such a disproportionate elevation in cardiac and hepatic tissue following shock suggests that ATP is selectively concentrated by these tissues during shock. Although both organs are drained of blood prior to ATP determination, neither is totally asanguinous. Coronary^{17, 20} and hepatic dilatation might account for some of the increase, but the more than 25-fold increase over normal dilutional levels makes this unlikely.

Other possible pharmacologic activities of ATP in shock include its activity as a cardiac glycoside¹⁶ or replacement of deficient high energy phosphate.

SUMMARY AND CONCLUSIONS

1. The protective effect of ATP administered both before and immediately following hemorrhagic shock in rats has been compared to analogous doses of AMP, DBZ, and pyrophosphate.

2. When administered 15 minutes before bleeding, there was 80 percent survival in animals receiving ATP compared with 30 percent survival in the saline controls, 20 percent in those receiving AMP; 12.5 percent in animals receiving DBZ, and 40 percent in those given pyrophosphates.

3. When administered following hemorrhage, there was 60 percent survival in the ATP group.

4. It is concluded that ATP has a protective action when administered prior to experimental hemorrhagic shock in rats. This action is not due solely to its vasodilatory action, since equivalent hypotensive amounts of both AMP and DBZ failed to afford an equivalent protection. There is a statistically significant beneficial effect when ATP is given after bleeding.

5. Since equivalent amounts of pyrophosphate and AMP do not afford such protection, it is concluded that the terminal high energy phosphate bond is responsible for this activity.

6. ATP administered prior to hemorrhage minimizes the depression of ATP tissue concentration in the liver of the shocked rat compared to untreated shocked controls.

7. ATP (6.66 mg. per kilogram of body weight), regardless of concomitant blood replacement, failed appreciably to alter cardiac output in dogs bled into shock utilizing the Wiggers technique.

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DISCUSSION

Dr. F. A. Simeone (Cleveland, Ohio). I have enjoyed this presentation by Dr. Eiseman even more than you have, because I had the privilege of receiving the manuscript beforehand. It is a very important paper because the work is representative of the excellence and precision which we have come to expect of Dr. Eiseman, and also because it bears on a very important aspect of the shock problem, namely the metabolic derangement in lethal hemorrhage. In this condition, the circulation has failed as a transport mechanism, and in proportion to the degree to which tissues are not adequately perfused with blood, energy metabolism becomes glycolytic or fermentative instead of oxidative. Aside from this fact that results in metabolic acidosis, the production of materials which yield energy for the conduct of life processes is very much decreased. For example, from a gram molecule of dextrose, glycolysis produces the equivalent of 36,000 small calories of energy as compared with 686,000 produced by oxidation. Therefore, it is reasonable to rest the possibility of protecting animals from the effects of poor tissue perfusion by providing them with energy yielding material, the universal and most important one being adenosine triphosphate (ATP) which the authors used, and which we also tested in our laboratories upon the publication of the authors' preliminary results. Any substance which salvages six out of eight animals which otherwise would have died of lethal hemorrhage is worth studying in its own right. Now, it would not have occurred to us to test this therapeutic possibility for a number of reasons. First, in order to be effective as a source of energy for biologic processes, ATP must exist in cytoplasm inside the cell. It does not function in the blood. In fact, ATP, although produced in all living cells, both plant and animal, is not ordinarily found in plasma, which is carefully freed of cells even with the most up-to-date methods for analysis which employ the tail end

of the firefly. Secondly, there is no evidence that administered ATP is capable of finding its way into cytoplasm across cell membranes, whatever these may be. Not only is ATP a large heavy molecule, which therefore would have trouble crossing membranes, but also it is a highly polar molecule, possessing three or four strong negative charges. And one would expect that ATP, like other highly polar substances, would be incapable of crossing the lining of a living cell. I suppose it could get in as a hitchhiker attached to a fatty acid, but to my knowledge there is no evidence for this. Conceivably, ATP could act in some beneficial way different from the pharmacologic actions of the other substances which Dr. Eiseman tested, but it is difficult to visualize what such an extracellular effect would be which would depend upon the highly unstable, high energy, low entropy terminal phosphate bond of ATP. Nevertheless, in view of the favorable results reported we, too, tested the effectiveness of ATP in lethal hemorrhage. In our laboratories, Dr. Mahassen Ghobrial performed the experiments, following the protocol described in the paper presented as nearly as possible. The experiments were all done in paired rats and supervised by Dr. Hopkins and myself, and our results seem to show that the use of ATP was not the determinant for survival or death.

We are at a loss to explain these differences in results, but believe it is a statistical problem. Certainly more work is needed by ourselves as well as by others, because a strikingly positive result such as that presented by Dr. Eiseman cannot be contradicted by a single negative experiment such as ours, and because this is indeed an important area to attack in the field of shock, namely the extravascular and intracellular circulation which we might call the ultramicrocirculation, concerned as it is with the transport of substances, including electrons, beyond the "microcirculation" across tissue spaces, membranes, and intracellular paths.

Dr. R. Adams Cowley (Baltimore, Md.). I would like to congratulate Dr. Eiseman for two reasons: The first is that Dr. Eiseman is working with a substance which can be given after shock has already begun rather than the use of the usual routine of priming the animal with medication before the shocking procedure. Second, because our discipline is beginning to think about enzyme systems and energy transfer at the cellular level. We have been working over the past four years using like substances such as mitochondria and electron transport particles. These substances, when given after the shock, begin to show promise. More effort is needed in these areas because, after the cell has been damaged, some type of support is needed to protect cellular integrity until recovery can ensue.